

Kushal Patel

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EDUCATION

Dwarkadas J. Sanghvi College of Engineering, Mumbai | India

Expected May 2028

- B.Tech, Artificial Intelligence and Machine Learning
- Honours in Immersive Technology
- CGPA : 8.89/10

PROJECTS

Solar Flare Forecasting

Ongoing

Vision Transformer | Sunpy | XAI

- Collaborated within a team to implement a custom **Vision Transformer** architecture with **cross-attention** across sequential solar frames to forecast flare occurrences.
- Curated and preprocessed an extensive **time-series dataset** consisting of **87,600 solar images** spanning over a decade of historical events.
- Integrated visual attention maps into the inference pipeline for **explainability**, utilizing spatial focus regions to detect **model drift**.

FieldSight Lite

September 2026

Computer Vision | Edge Computing | CIELAB | Statistical Modeling

- Engineered an **unsupervised, training-free** vision pipeline on low-power CPUs, achieving **35.8ms latency** (27.9 FPS) with **68MB RAM usage**.
- Implemented **CIELAB color modeling** and **MAD outlier detection**, outperforming classical baselines with an **83.35% Foliage IoU** and **+71% relative gain** in lesion detection.
- Built an optical stress harness across 5 lighting regimes, maintaining a **62.3% LRS** and bounding severity drift to just **3.07% MASD**.

FirSeFile

September 2026

Python | Swin Transformer | PyTorch | ONNX Runtime

- Collaborated on a forensic recovery platform, integrating an ML carving pipeline into a Rust backend to process raw disk images.
- Built a **Swin Transformer V2** model achieving **~93% accuracy** in classifying orphaned, headerless 4KB blocks across 10+ file formats.
- Exported to **ONNX Runtime** to achieve **<8ms inference per block** and used **probabilistic graph reassembly** to reconstruct non-contiguous fragments.

SKILLS

Languages: Python, C/C++, SQL, HTML, CSS, JavaScript

AI/ML: PyTorch, Computer Vision, XAI, Machine Learning, Deep Learning, Scikit-learn, Pandas, Numpy

Web: Next.js, FastAPI, Flask, Django

Tools: Git, Github

Achievements

Winner – Lines of Code (LOC 8.0) 2026 National-Level Hackathon out of 1000+ participants

Collaborated on Exportify, designing the multi-factor scoring algorithm and trade-matching engine that ranked exporter-buyer compatibility across multi-dimensional features to automate international supplier discovery and trade risk assessment.

1st Runner-Up – Drishti AI Hackathon 2026 organized by Dexit Global out of 250+ teams

Collaborated on CopyCop, designing the behavioral risk-scoring engine and multi-person tracking pipeline to detect examination misconduct anomalies and stream real-time, privacy-preserving alerts to invigilators on edge hardware.